

DSA 8405 Probability and Stochastic Processes-20250924_163011UTC-Meeting Recording

September 24, 2025, 2:29PM

2h 1m 7s

OC **Odhiambo, Collins** 2:12

Uh, hi Cosmos, could you reach out to the rest of your colleagues, see if we they can join?

CK **Cosmas Kungu** 2:19

Thank you. Thank you. Let me reach out to them. Thank you.

OC **Odhiambo, Collins** 2:23

OK.

OK. So I think we can just proceed. I don't want us to take long for this because I just want to introduce a bit of a concept on birth and death process.

And then?

RJ **Rutendo Julia** 8:44

1.

OC **Odhiambo, Collins** 8:46

Uh, see.

Yeah. So you introduce the concept of bath and death processes and then we can derive some few equations.

We're almost there. I've already set your exams. I think for now there are only a few pieces that you probably require for you to do your exam.

Um.

Though we still have some topics to cover. So the exam, the exam is pretty set for now though, but we still have some few topics. So deliberately I will teach those topics, but of some of them I've left them out of.

Testing because I find them a bit abstract at this level, but just to introduce it so that you don't get stuck when you some of you decide to come for your PhD and in your PhD you're required to do some.

Uses on stochastic processes.

So, but just that said and done, I will share with you like a practice like something like a practice question just to help you prepare well for your exams. I think your exam should be somewhere around.

Around next month. Is it April or it's not April, it's October.

And then it's October or November.

CK Cosmas Kungu 10:23

It is the early week of early week of November, the first week. Yeah, and we are praying that you look us with good eyes. The semester is so heavy.

OC Odhiambo, Collins 10:27

Oh, I'll wake up. OK, OK.

OK.

OK, that's fine. I will. What I'll do to help you out is probably prepare for you some set of practice questions so that you have sort of an idea how the exam will look like.

Yeah. So I think that's that's how it will be. The exam will be 60%. So these cuts we are doing will be 40%. So if we try to at least submit all the cuts and.

Get something close to 3030 / 40 overall after conversion. Then I think for exam you need you need at least 40 out of 60 for you to score an A, which is possible in this course.

Yeah, I think it's possible to get an A several of you.

So today I want to look at the pure, the birth and death processes. Sometimes we call it your refuge for birth process. So these are two processes. So there's the birth.

And there's a death. So if you combine is when you get birth and death process, OK, so generally.

So, so, so generally, uh.

That this is just by name intuitively, but practically you can apply in any scenario as long as you can define your birth process as some arrival aspects. OK, and then death is some exit, but now we consider it in a special way, not like.

The queuing system where we have arrivals, so we have the arrival pattern poisson and then the exit pattern we call it service mechanism, which we assume it follows an exponential, naturally an exponential.

Distribution. Now this looks like the same scenario, but we put in some few aspects so that you look at it's like you tweak arrivals so that you look at that as birth process.

And then we tweak exit and we look so that we look at that as a death process, OK. So just from a basic scenario, you could look at stability, people, demographers, people studying population stability will be interested with this kind of process. OK. And look at fertility rate, the rate of fertility. So that's every birth is a birth, fertility area arrival as a facility is an arrival process, is a birth process and then death is an exit.

Yeah.

Even in the migration, um, you could look at.

Immigrants, immigration as birth and then immigrants as death. And so now you try to stabilize the system so that you have arrivals that that has certain characteristics and then you have exits.

Uh, that defines us death processes.

OK.

It has a bit of relationship with Markov. I will not go into the mathematics a bit, but we'll try to see how best we can model your birth in that process. So for now I think the Kolmogorov equations.

The Kolmogorov equations should be should be able to work. If you remember the Kolmogorov equations, that should be easier for us for now.

So that we come up with those set of equations and then once we have an ordinary differential equation then we can model it.

So the Ulfuri process or pure birth process, you have to consider a cell which produces according to the following rules. So it follows these certain rules. So a cell present at time T has a probability of.

λH plus this small value here is a negligible small value that tends to 0 faster. So you understand that H is a small value.

λH is also a small value, but this λH attends to 0 much faster than H so that if you introduce the limits then you have the limits of λH over H as the H tends to 0 then you end up with 0.

OK. And then we split the intervals into that interval of H plus that and then the probability is independent of H .

OK, so if you're looking at cells, the way cells reproduce themselves and this one we do pretty well when we are doing what we call in host modelling. When you are you are, we are doing in hosts modelling in biomathematics.

You can model it and come up with what we call the reproductive factor. OK, so something like coronavirus, COVID-19 virus.

So you only need one cell or two cell, one cell to find its way into the lungs of someone and then it begins to multiply.

And the amount of multiplication, the way it's multiplying with any small change in time is independent of the type of of the age of the virus.

So that the event between different cells are independent. So you start with one cell, it's multiplies, produces 2.

This one generates 2. This one also generates 2 and probably it dies here. This one generates 2. This one takes long before generating and then the one one of the child's dies there, but the other one generates too. So you have a system.

Like that. This is now what we call a pure birth process and this is just an example. So you start when they do birth, when they do fertility rates, they don't look at, they don't look at male, they look at female.

Yes, it's only few women who we can use because they are the ones who have the ability to reproduce. Of course they also need men, but when we do fertility rate, we look at women. OK, so so one woman who is in age.

Reproductive age. It's um can produce two or three or even just one.

So probably after two kids are born, one produces 2, the other one refuses to give birth, or the other one is a male is now not part of the chain.

But the other one is a female. So something like this so that you have a chain of reproduction over time and and this is what we call a pure birth process. You will see for death processes. Let me just jump quickly to go to death.

You'll see if you if we introduce the death process then.

Apart from giving birth, we have generation which are coming in. So this first component here will be a generation that is coming in and this is the generation that is exiting so.

In the so you have generation that is working in and the generation that is exiting. So this exiting is exiting at certain rate. So you have now have a death process. So if you measure birth and death process now you can model.

Population stability in that case. OK, so that's that's just not that just let me not go there further. Let me go back.

Um.

To the birth process, the Ulfuri process. OK, So what we are saying within a short time interval, the the probability that there is a cell that will a cell will be generated within a a specific age interval.

The probability is this.

And then each age is independent of age, the probability is independent of age, and then each cells events within the cells to the cells are independent.

It.

OK, so there are two approaches. So there's a non probabilistic approach, which is this. I find it much easier. Let me try to connect it to the white board so I can explain well.

OK, so if we say that N_T is the number of cells at time T .

Then the number of births, the births that occur between this age period is N_T is λN_T times the small change.

Where λ is the birth rate per single cell.

OK, λ is a path rate per single cell.

So whatever happens between this time is this one. Let me let me just connect it with the whiteboard here.

OK, I just want to show us.

Some little bit of that computation, but basically what we are talking about, we're talking about a difference, what we call an ordinary differential equation, because if this is the number of cells at time T , this is the number of.

Cells at time T So the number of cells at time $T + \Delta t$ is the number of cells at time T plus the interval.

OK, plus the number of cells in the interval.

SK **Stephen Korir** 22:37

Hi, Prof.

OC **Odhiambo, Collins** 22:39

Yes.

SK **Stephen Korir** 22:40

Just a clarification, I came to class a bit late. When is when you talk about the number of cells at this point, what do we, what are we referring to?

OC **Odhiambo, Collins** 22:52

Uh.

So here you started at one, so you started with one. So that's that one after some time it generates like 2. OK, so because it's a bath process, so there's.

SK **Stephen Korir** 23:07

Mhm.

OC **Odhiambo, Collins** 23:11

Continuous birth, assuming everybody that is born eventually gives birth. So we look, look at, look at cells. So that cells over time it splits into two and then over time each of those two have matured, split again.

SK **Stephen Korir** 23:18

Mhm.

So, so, so, so the cells here represents an entity which is a person or.

OC **Odhiambo, Collins** 23:33

OK and.

Yes, yes, let me put it this way. For proper comprehension, it's much easier to look at it from a cells part rather than a human, because when we look at human.

SK **Stephen Korir** 23:55

Mhm, mhm.

OC **Odhiambo, Collins** 24:00

It's only male. We consider male, female as the only people that reproduce. Of course there's a there's a contribution of male, but when it comes to fertility rate, we only consider female, so.

SK **Stephen Korir** 24:07

OK.

OC **Odhiambo, Collins** 24:17

You could probably start with a daughter that is born. So you start with one daughter. When that daughter matures, generates 2 kids, two daughters. So assume here you block off men, block off men.

SK **Stephen Korir** 24:22

Mhm.

OC **Odhiambo, Collins** 24:33

So after a while, each of those generates 2 children. Some will decide to remain without giving birth, but some may take longer or get give birth at 40. Some will give birth earlier, like this one.

At 20. So they what we're saying that duration is independent. The longer the duration takes is independent. But now if you assume that none of those daughters die, so the number of cells are all of them. So overtime at time T .

You'll count everybody in that ecosystem, so that becomes like most now if you look at the whole number of sets.

SK **Stephen Korir** 25:15

OK.

OC **Odhiambo, Collins** 25:15

That that's now you are NT because in a stochastic process again, so we have as time as time progresses as. So the parameter time determines the state. The state is now in this case is the number of cells.

In in in this kind of setting that we are looking at, yeah.

SK **Stephen Korir** 25:41

OK. Thanks, Prof.

OC **Odhiambo, Collins** 25:43

OK.

So.

If if you consider at time T , if you consider at time T , that is after, let's say after 100 years, how many children would be there? So assuming nobody dies, whoever is born.

Lives forever and eventually has the potential of generating a new person or a new cell. So if you look at cells, it's easy to think of a cell because of like bacteria. That's how they grow all viruses or.

Whatever this microbiological elements, that's how they grow. They grow, they they after they mature at some point it split, they split and then each becomes a descendant. So each grow again and splits and each grows and splits over.

For us some time so.

So if if you look at the number of cells at time T , that will be your N_T . Now if you look at number of cells at time T plus small change like an H , this is a small change.

The number of cells at time T plus that small change is actually what you got at that time N_T plus the small change, that small change and that small change. These are the number of births that are of cut, assuming that the rate of birth per cell.

Is λ and then you multiply by N_T times the small change. So that's that's where that's where we are going.

OK, so if I just try to, let me just try to see if I can move.

To.

The white bone.

OK.

OK. So we're looking at this is the time or the the time. So we started at time 0, so we have $1/2$. So these are the time periods all the way up to sometimes T .

So at every time, after every time the number of cells split or generate new cells, at every time they generate new cells. So if you accumulate each now at every time, if you look at the number that you have accumulated, that becomes your state, your space, the state space.

So that at after the at the end of it then you have N_T as the number of cells at time T .
OK.

At time, at time T So you have that number of cells at time T So what we are saying is that if I take, if I consider T plus and here let's say there's another.

In in between here is a very small gap which I'm calling ΔT or call it H / L So if I take, if I take, if I look at the number of cells at at this point here, here.

Yeah, is T plus that small change.

OK, so if I take the number of cells at at at this point is the same as the number of cells from zero up to here at T plus in between.

So here is the same as the number of cells at in between up to from zero to T and then now plus the number of cells in between. OK, in between T and that small change.

OK, So what we are saying that this one is defined from pure bath process. This one is defined as λ multiplied by the time, the number of cells from here to here.

And then you multiply by that small change.

OK, So what we have then is Δy plus the small change that will give us Δy .

Plus $\lambda \Delta y$ that small change.

Right. So if you bring the, you bring this Δy to this other side, you'll end up with Δy plus. So you'll end up with Δy plus plus that minus Δy .

This this is the same as $\lambda \Delta y$.

And the small change. So that's what we are saying. And then now you can divide both sides by Δy by that small change.

And then introduce the limits as that small change tends to 0.

So as that small change tends to 0 of Δy plus λ plus Δy .

No, no, I mean, just try it again.

So if you introduce a limit as that small change tends to 0, then we have Δy plus λ .

And that's in to.

λ^2 is equals to limits and.

This is not λ this so because you introduce the limits on both sides. So this is Δy .

Δy .

Are you together? Can you guys? Are we still connected?

Are we still connected? Just confirm. OK, OK, OK, OK. So so that's how we do. So we divide and here we had of course we had Δy . So you what we've done we've divided both sides by the small change and then introduce the.



Elias Shamalla 33:10

Yes, yes.



Odhiambo, Collins 33:26

The limit on both sides. So this cancels that the limit of a constant is still a constant.

And then this one here, if you remember when we were developing Poisson, this one here is differentiation by first from first principle.

That's differentiation from first principle. So this is the same as BDT of Δy .

And then this other side is λ . Um.

But I'm done.

Δy .

Yeah.

Yeah, so that's now we are left to solve this.

This is now an ordinary differential equation.

So we are left to solve that ODE. So the best way to handle that is to bring make the constant one side, so bring the functions together for those of you who have done ordinary differential equations.

This is the same as. So if I do DDDT, let me just use it in a in a more general way.

And then down here I have.

And T and then this is the same as Lambda.

And it's sent this back.

All right, so who can? Is there anyone who can tell me how to solve this?

And this is an ODE.

Yeah, like more general, I don't know if Bernard you you can think about it because this is the same as N the derivative of that over NT.

Or anyone who can try to help me think this. Of course I already know what we need to do. I just want to see if some of you.

Have or can recall some of the things you did in under in first degree. For those who did either double maths, applied maths or pure maths, you must have done ordinary differential equations.

Anyone who has an idea how to move around that?

Uh, Bernard, can you try?

Alright.

OK, so let me explain how to you can solve it. Uh.

I think the easiest way to help you see what I'm going to see to do is to take you back to differentiation, just quickly take you back to the concept of differentiation.

How many of you can do for me this?

5.

This will be what?

From using power rule, what will be that?



37:08

4.



Odhiambo, Collins 37:08

That's a normal differential.

SK **Stephen Korir** 37:09

Type X raise to 4.

OC **Odhiambo, Collins** 37:11

What? What?

WO **Walter Ouma** 37:12

It's powerful.

OC **Odhiambo, Collins** 37:15

Perfect. Uh, what about? What about this?

A.

SK **Stephen Korir** 37:28

$1 / X$.

OC **Odhiambo, Collins** 37:30

Buffet.

One of X. What if we have a function within a function? Let's try a function within a function now here. What if you have?

Three $X^2 + 3$.

Our What will be that?

SK **Stephen Korir** 37:56

It will be $3 * 6$.

OC **Odhiambo, Collins** 37:58

So here you MP3. So you you do power, you do function chain rule, chain rule. So you do chain rule. So this will be $3X^2 + 3^2$ multiplied by.

The function inside the derivative of the function inside, which will be $6X$.

WO **Walter Ouma** 38:19

$6X$.

OC **Odhiambo, Collins** 38:21

So what you have done is that you've differentiated the outer function and then multiplied with the derivative of the inner function. OK, now what about this one now if I do DDX?

LN um.

3X squared.

What will it be that?

SK **Stephen Korir** 38:57

It would be.

$1 / 3 X^2 * 6 X$.

OC **Odhiambo, Collins** 39:05

Yes, multiplied by 6. Perfect.

Uh, who is that? I need to send you. Who is that?

SK **Stephen Korir** 39:16

Media.

OC **Odhiambo, Collins** 39:17

Who is that? That's who I can't. You know, I can only see the the the the short SK. Who is that?

SK **Stephen Korir** 39:24

Uh, Stephanie.

OC **Odhiambo, Collins** 39:27

Please send me an e-mail with your with your your Imperson number. I need to. I need to send you a small lunch.

SK **Stephen Korir** 39:38

Mm-hmm. Uh, thanks. Thanks.

OC **Odhiambo, Collins** 39:38

But tomorrow I'll.

OK, so uh.

So, so that's that's how. So what what what Corrida has done is that you've taken you've done the derivative of the outer function which is one of what is inside multiplied by the.

In a function, the derivative of the inner function here which is $6X$. So generally for LN if you are to do DDX of LN of a function and there's a function inside that LN.

Is the same as you do one over that function.

Multiplied by the derivative of the inner function, which is that which that will give you. So the derivative of the inner function is that.

Is that and then you have that function itself is.

OK, so that's if that's the case. Um.

Look at this one now. Just look at this now. This is what we are dealing with here.

So if you look at this one keenly, in order for you to unpack this, you need to think the reverse that this is already a differentiated version. So can we go back? So for example?

Here we are able to go forward. If we're given LN of F of X , we can differentiate and write it this way. But if we're given this form and you go back, write it back into that form.

OK, so if you look at, if you look at this expression here, it is in a derived form. It's like the end product. Now can you reformulate it back before it was it was differentiated to end up the way it is?

So if you if you think that way, then it's much easier to work on this because this one then means that that that means that we have DDT of LN.

And.

T.

Is it making sense?

OK, so so so you have to, so you have to think. So this is just calculus too that is helping us solve that problem. It's just calculus that's helping us solve that problem.

It is our ODE.

A very simple deal, but if I'm thinking from a non probabilistic format and just look at it from a high class point of view.

Then I have to think backward. Can you reformulate backward and and when you reformulate backward, can you see that this is exactly what we are ending up with?

OK, yeah.

Right. So that's that's now what we're ending up with. So now we we are left to solve this. We are left to solve this. Now it's easier to solve this because now what we need to do is just to multiply across or rather put this operator on this side.

So we have DLN of NT . That should be the same as ΛDT .

Then if you integrate both sides, integrate both sides, this side will be LN or just let me just put it in full log of ELN is the log of E .

NT is equals to Λ .

T .

OK, so just rewriting. So the rest is just algebra, just rewriting. This is the same as $E \Lambda T$.

It was.

Of course, if we integrate, yeah, yeah. So that's the reasoning I wanted you to see.

That's the specific reasoning that I wanted you to see.

So as we look at the pure bath process, but but you you should just see clearly that in most of these stochastic modelling, the Kolmogorov equations, the Chapman Kolmogorov equations.

It could be forward or whichever. Some people call it forward equations, whatever plays a very critical, plays a very critical role in modelling.

And in fact, if you are very interested in mathematical modeling like biomathematical modeling.

I don't know about opportunities currently in the country, but you know we don't do a bit of rigorous research around some of these, but in the West these are very important areas to really look at.

Laboratory resents.

They are funded. Some of these laboratories are well funded. I think in Africa one of the one of the factors we miss is research funding. We we we don't invest in research and such our government don't put.

Money in research as much as possible, as in the West and so.

The application of these things becomes important, especially in genomics. OK, so genomics, where you want to manipulate genetic growth or composition either because you want to solve a disease.

Do you want to come up with a cure of a disease or look at what we call drug resistance in the body? So most of genomic genomes or genomics where?

We model, we model using some of these processes is are very much useful when you want to come up with treatment of a disease or look at what we call drug

resistance.

Drug resistance or many other. There are many other things you look at. OK, so just think of it like.

I can only explain it well in HIV because I've done, I've done more years in HIV, but I'll think of it this way in an HIV setting that the fast infections the.

The first infections that get into the body, what what we call.

That's the first, the first virus that get into contact to the human body. Call it wild virus. That virus is called wild because it will seek to survive inside the hosts inside the human body.

So and then it will once it's it usually settles after 72 hours is when it will settle. That's why there are some interventions that needs to be given to the to anybody.

Some prophylaxis call them post exposure from prophylaxis.

Which we you you give it to someone to avoid to prevent the virus from getting old or settling down there. OK, so interventions like gender based sexual violence.

People who have become or rather survivors of sexual violence. The first thing is within 72 hours they should be put on pre pre exposure or post exposure what we call pre post exposure.

Prophylaxis so that we prevent in case in the event there was a virus, then that virus cannot settle. We are able to arrest that virus within 72 hours. Now once the virus settles, if it manages like in the early years of HIV.

It settles within. After once it settles, it begins to generate children to split. See the way we are looking at the cells, the birth process, the cells begin to split and split. So it is wild virus because it's splitting at a very high rate because it wants to survive, so it has to generate many kids.

Or many descendants at a very high speed. OK, the first three weeks. I'm just trying to explain to you how the modelling is done using some of these concepts.

So the first the first three weeks, the body will realise that there's a there's a there's a stranger in the house. OK, so your the body will will realise there's a stranger in the house. So what the body will do, it will match all it's all resources.

To fight that stranger who is the virus.

So unknown to the body that that virus has already settled in a place where it cannot be removed. So the body will generate a lot of CD4 cells and CD8. Those are cells that fight diseases. OK, so these are cells that fight diseases.

Now if that person goes to for HIV test during the first three weeks, it will not show because there's the body has already generated enough white blood cell, the CD48,

CD4, the CD4 and the CD8.

To fight the virus. OK, so the that person is so much healthy that time because the body is trying to fight a stranger who has come in, who has knocked in and come in. Right. But after 345 weeks, the body realises, uh, this guy can't be kicked out. So we have to negotiate because if you try to kick him out, he's already settled in a place where you'll need to destroy the entire cell so that.

OK, so we have to negotiate. So the body now begins to negotiate. OK, you you are here. Yes, we've tried to fight him. We can't fight him. So we have to agree how to survive because the body also has to.

Has to survive, right? And the virus has to survive. Now with time that virus, after it has generated a lot of children, the kids also have grown and begins to compete with the parent, so they even try to.

They also endanger the life of the parent. So you have the wild virus and then you have the descendants of the virus and then you have the body.

You can only present that mathematically using these equations. OK. And then of course over time the immune system is compromised when the virus have increased in number, so the immune system is compromised.

And once the immune system is compromised, some opportunistic infections begin to shore up.

Like TB or hepatitis, typhoid, some very some opportunistic infections were just waiting to to shore up. So once they tried started showing and the person goes for a test and then it is discovered that.

He's suffering from what we call a human immunodeficiency syndrome, acquired, we call acquired immunodeficiency syndrome. So this is something that will be there for a long period of time. OK, so the.

Clinician or the doctor puts the patient on ARV. Now ARV. When ARV comes, then we introduce aspects in this equation because now we have a birth process, but if you introduce, so this is a birth process, but if we introduce ARVS.

Then we are diminishing the right of path.

And so you have, so you introduce another element and that's now what we call the in-host modelling. OK, and those elements now you can solve using ordinary differential equations. We can now.

Try to solve it. Now you can look at.

The rate at which virus are being destroyed. Of course you can't eliminate all the virus.

You've introduced the drugs, they are used to limit the rate of growth. OK. So if the reproductive rate here was Λ , then there's a factor we introduced that lowers this Λ and that is an effect of an ARV.

OK, so once you generate all those equations using the Kolmogorov, then we can look at the longevity, how long it will take before eventually the virus become immune to the drugs because over time the virus can develop what we call resistance.

Especially if somebody is not adhering to the drugs. Yeah, so, but in a nutshell, that's. Uh, that's just uh, an example of um.

What we call a pure birth process and that's what I wanted to show us here.

If this is the number of cells at time T .

In between the cells, this is the number of cells. So if you're looking at T plus at time T plus that small change, so it is the number of cells in that plus this and then of course.

If you bring this and introduce the limits, then you have that and I've shown you that this one is an ordinary differential equation which ended up with this. So if you integrate both sides you end up with.

That and of course that is the value you have and then the small changes. This is the value at N naught at time naught. So it's more of what we've tried to work on.

OK, so uh.

Looking at it is this format, so we've already worked looked at that.

Let me see what is happening here if the small change. So this is now when you look at it from a probabilistic point of view, but if you look at it from an unprobabilistic, we got it here.

So this is the probabilistic. OK, let me just go here. So if you look at it from a, so a non probabilistic is what we have done. So if you are, if you approach it from a probability, it's the same way.

It's just the same way the way we looked at Poisson because it is an arrival, so you look at it from an arrival point.

OK, so this guy called Julie, so the guys who came up to make some of.

The concepts that we are applying, he studied this process in connection with the theory of evolution. The population consists of species within a genus of creation of a new element.

Is due to mutation, so negligent probability of species dies out and the size of the species. So and then there was another guy who came up with few, it's called Fury.

So there were there's Julie.

Then there's Fury. We use the same model when he looked at radioactive mutations.

So he may look at it from a geometric point of view.

Um.

So a geometric or a geometric model probability. We model events when you're interested in the number of events that have occurred at time at a certain time period, OK.

So usually for geometric, geometric, we are interested with how many successes, no time, how, how it takes to the how many number do you have to reach your first success, OK and to the first success.

So you have failed where Q is a probability of failure, P is a probability of success. So you have failed, fail, fail until you get a success. That is what we have a geometric progression.

OK, so of course this one is using probability generating functions, so I'll not go into that. That's when you're applying probability generating function.

But this is still important. We'll come back to.

Uh, and see how it branching process.

Looks like.

I disappeared. I'm now back.

OK.

Right, so we will still come back to a bit of this. Basically geometric. This is a geometric generating function. I don't know. We can cover it probably after a short break. I'll I'll explain to you how it.

Metric progression. This is important because we will still apply some of these concepts when we come to the what what we call the branching processes.

Yeah, so the two eventually merged and then we have a pure bath process. So this one is using geometric, it's called probability generating function.

All right, so let's take a short break of 10 minutes and probably I'll take us through the the geometric, the probability generating function of this geometric progression because we still need, we will need it in branching processes.

So let's take a short break of like 10 minutes.

Alright, so um.

Let's continue.

A bit.

Right, so let me just.

Um, take us shortly on.

The white boat.

Uh.

OK, so it's just a facility that I want to show us a mathematical facility that is important when you are doing some of this modelling, especially if you have a discrete sort of event. It's called probability generating function.

Sorry.

OK, so the probability generating function is 1 facility we have if you probably are interested in estimating events in your sample in your population and.

The probability the the probability distribution is undefined or unknown, and sometimes even if you know the probability distribution, it's much easier to work with generating functions. I know you have done.

Uh, let me just get a sample from you people. Have you guys done anything like? Moment more I've ever heard of this.

Looks like I disappeared.

Um.

So have any of you heard of something like this? Moment generating functions? You must have had it. Let me just get a sample from.

Um, anyone who has ever had of moment generating functions?

That should be in a classical probability class and some advanced units in probability like.

By the time you're in third year, if you're doing something, you're majoring in statistics.

Or.

Yeah, let me just see. I'll get it from you some a sample from you guys. Anyone who has ever dealt with moment generating function or if you ever had moment generating functions?

BO **Bernard Ogol** 1:17:52

Yeah, I think I've done them.

OC **Odhiambo, Collins** 1:17:56

OK, so that's bad. Anyone else just need to have quick sample?

OK, OK. So at least we have a sample of two. So if I assume the rest, at least you you may not have covered, but for this class, for this course, we don't need the.

We only need.

The PGF probability generating function, some models, some stochastic processes like branching process. You can easily solve some problems using.

Probability generating functions. It's much easier to work with probability generating function for some of those probability, some of those stochastic processes. So like the birth and death process. So so let me just assume.

Most of you have no idea about it and just give you a basic foundation around it and then we'll see how it is applying in that pure path process. OK, so the PGF, the probability generating function, I could call it PS.

Is defined as you do you sum all values of so all values of X . So introduce S^X into the probability distribution.

So you introduce. So what it means is that you sum. Remember this is basically and it's only used when you have discrete.

Uh, probability distribution.

So any anytime we have discrete, that's where you can apply. Now MGF you can apply it whether you have continuous or discrete PGF. It's only applicable when you have discrete probability distributions.

Example of discrete like Poisson. Of course discrete distribution applies where the number, this whole number is a natural number.

So so you have the the the random variable can only pick what we call natural numbers.

+ 0 that means 012. So it's natural number plus probably 0. Zero is not a natural number, but so natural numbers as you know it will be numbers that cannot assume. Pessimos all fractions.

OK, so numbers are cannot assume decimals or fraction.

OK, so so if you have those numbers, so so the probability generating function is applicable when you have those events that assume discrete probability distribution.

OK, so for example, which is the easiest? So for example, uh, assume.

So this is just a simple example just to try to build some foundation to you on probability generating functions. So assume we have this discrete event, so X can pick values 01 and two.

With probabilities probability that X speaks value 0 is 0.2. This is just an example.

Probability that X speaks value one is let's say 0.3.

And then probability that X peaks value 2.

In 0.5. So this is just an example of a discrete case. So a discrete because X can only

pick 01 and two not in between can only pick 01 and two and not in between and therefore with the following respective probability.

So.

If you want to get the mean of that probability distribution, it's very easy. So you just do the sum X moves from zero to two.

Then you do introduce a factor there and then you multiply by their respective probabilities.

So the mean would be $0 * 0.2 + 1 * 0.3 + 2 * 0.5$, which will give you.

Is it 1.3?

And then you can also get the variance.

Of X for that type of for this type of. So the variance is the same as the the second moment multiplied by the square of the mean. So this part here is already known as 1.3. We got it here.

So we only need to work on this part and then we subtract and get the variance. We need to get to work on that part and that part is $0^2 * 0.2 + 1^2 * 0.3$.

$+ + 2^2$ times 0.5 their respective probability. But this time you're squaring first times their respective probability. That gives you this moment, second moment and then there you have $0 + 0.3 + 4$.

Now that's that's two, so that now that part is 2.3 and therefore the variance of X is $2.3 - 1.3^2$. OK, so that's just the basic.

Way of getting the variance of a very simple discrete case, discrete probability distribution, my mean and variance. Now you can. There's another approach you can use that is not this approach. This approach here is generally called method of moment. This is a classical way of of getting.

the mean and variance of our discrete probability distribution. Now there's another approach which you can use, and that approach is called probability generating functions.

So you can use the approach of probability generating functions to get the mean and variance. So it's just another way of getting the mean and variance of estimating the measures of that probability distribution, but using a different approach which we are calling probability generating.

Function. So the PGF is defined as below. I've I've defined PGF here, the probability generating function. I've defined it there. So I just need to plug in the distribution and then work on that function.

And then the mean is you get the first derivative, the the variance you get a second

over N OK, so the PGF, the PGF.

Or what we would call PS is basically when you when you sum all values of X by you introduce a function $S^{\text{power } X}$.

Into the question.

Into their into their function.

Yeah, so if you want to get, if you want to get the mean.

You differentiate, you differentiate with this respects to S , then you put S to be one.

So to play together mean which is the X is you do the DDS of PS and then after differentiating you put S to be one.

That would give you the mean. So that means then that if you differentiate this with respect to S this.

This will be if you differentiate with respect to S so the so the DDS of of this.

Oh.

So the DDS of that.

It will be will be. So you bring the X up because we respect differentiate with respect to S So this will be. So the summation will remain, the operator will remain.

And then you have $S^{\text{Yeah } X}$ here. Then here you have $X - 1$ and then here you have probability of X .

So after you've differentiated and then you put.

S to be 1. So if you put one here, you'll have $S^{\text{power } 1 - 1}$, which is $S^{\text{power } 0}$. So you'll end up with. You'll end up. If you put S to be one after differentiating, you'll end up with this.

Which is the mean? Which is this the mean?

So the mean is just get the PGF and then differentiate it with respect to S and then after differentiating place replace S with one. That will give you the mean of that probability distribution.

Using the PGF for you to get the variance of that distribution, then you differentiate twice, then you add. So if this is the first derivative, if the first derivative yielded this.

So the second derivative, the second derivative will be you bring now this and then you subtract by one. So the second derivative will be $^{**} -1$, so it's.

s to.

So if you differentiate with respect to S , that's that will be the 2nd derivative and then you replace S by 1 power. Anything is 1, so you'll end up with these. So then you can manipulate these by adding X and then.

Subtracting. Now you can subtract the the EX squared.

So the the the so that that's that's what we will work. So because this will be now if you bring in that will be that minus that. So you'll add, you'll add.

Let me see how that will work. I'm just trying to see, make sure and so that's that's so it will be S OK. So the mean using the PGF, the mean is basically.

So if I you find the DDS of of that probability function and then after getting the DDS you replace S with one. OK, so the same way.

Uh, variance.

Whereas you do, you do, you do, you do DDS.

Twice.

And then replace S with one. So you do DDS, you pass, so you differentiate twice and then you replace DDS plus that one. Then you do plus.

Plus they mean.

And then you subtract the squared.

So that's that's how you get the mean and variance using PGF. So if I try to come here.

Um.

So if I come, if I try to come here, let me try to use PGF to solve the mean and the variance of these. OK, so here I had a situation where I X contains 01 and two.

Probability X speaks 0 is 0.2, probability X speaks one is 0.3 and then probability X speaks two is 0.5. So that's what I had. So if I want to get the mean.

And the variance using the PGF technique.

So so I have XI have X can assume 01 and two and then I have probability X assumes 0.

0.2.

Community X.

I seems one is 0.3.

And then for BPX to.

So to get the PGF for that I need to sum.

In this case, the probability is running from zero to two, so I introduce S here and then I have now the probability, the probability aspect.

OK, so if I replace if I unpack this, I will have S power zero probability.

X is equals to 0 plus S power one probability.

Yeah, ** is equals to 1 + S power two probability 2X is equals to 2.

So this one anything power zero is 1, so probability that X is 0 is given as 0.2 S power one is S probability that is given by that. Let me write it to this zero point.

Maybe it's this.

Repeat this way and $0.2 + 0.3 S$ plus that is $0.5 S$ square. So that's the PGF, that's a P. PGF of that function. So for you to get the mean, you partially differentiate with respect to one, then you replace with respect to S , then you replace S to be one. So the mean.

I mean this one with respect to S , so this there's no S here, so that becomes zero.

This respect to S will remain 0.3 plus this respect to S^2 times here. That would be one. This will remain S and then now replace S to be.

One so that now you have 0.3 that's one that's give you 1.3 .

Which is the same as what we got when we did it using the other method. So we got 1.3 , we got, we got 1.3 here. OK, so if you use a PGF we're still getting.

The same answer. So PGF is just a facility, a method, a more an approach that you can use. So this in this case it doesn't look, it looks like the other one is much easier, but you can be faced with more distributions that.

Are not are a bit complicated. All right. I didn't want us to take long for this one, so because we are still meeting tomorrow, let me just stop there and not bother you a lot with the mathematics now that.

We're also meeting tomorrow, so we can finish up the other part tomorrow and then.

Um, yeah, let's see how it goes.

Right. So you enjoy the rest of the evening.



Elias Shamalla 1:35:48

Okay, thank you.



Straton Amodora 1:35:52

Thank you.



Cosmas Kungu 1:35:54

88.

● stopped transcription